

AVERY REYNA

(954) 662-3927 | avery.reyna16@gmail.com | avryryn.io | Washington, DC

EDUCATION

University of Central Florida

Bachelor of Science – BS, Social Sciences

Orlando, FL
2023

RESEARCH EXPERIENCE

University of Kentucky

Independent Researcher | Advisor: Jonathan Powell

2026 – Present

Lexington, KY

- Designed and developed the selected prototype for a machine learning-powered web application assessing global coup risk, establishing the technical foundation adopted by a multi-pronged research team
- Led a cross-functional team of 5 CS students in building new features for a full-stack ML web application, owning technical direction, task delegation, and code review

UC San Diego

Independent Researcher | Advisor: Brian Hempel

2025 – Present

San Diego, CA

- Led mixed-methods research project studying experienced developer-AI programming workflows, designing observational studies and a large-scale survey to analyze developer strategies and goals when leveraging agents to develop software
- Developed Python-based GitHub GraphQL scraper with parallel processing to extract over 2.5k developer emails from thousands of AI agent and agentic coding repositories for survey recruitment purposes
- Extracted actionable insights from expert developer interviews by systematizing prompt transcriptions and identifying recurring patterns in how experienced practitioners structure queries to achieve quality standards with AI coding agents

University of Central Florida

Undergraduate Researcher | Advisors: Amy Donley, Pamela Wisniewski, and Jonathan Powell

2021 – 2023

Orlando, FL

- Developed a novel data and machine learning-based early warning forecasting platform that estimates the monthly risk of illegal leadership turnover and overthrows of government for 201 countries across the globe
- Designed and implemented data pipelines using Pandas and NumPy that transformed raw coup data into visualizations used for internal reports and articles for online publications with over 500k readers a year
- Qualitatively coded 50,000 Instagram direct messages to create an ecologically valid dataset using human-centered design principles to provide insight into teens' social media interactions and train models to detect online risks
- Led an independent machine learning project that comparatively analyzed two text summarization models in order to test its perceived utility for data annotation and qualitative analysis of large datasets more broadly

XR Access

REU Participant | Advisor: Brian Smith

2022

New York, NY

- Designed and conducted nine usability studies of two web applications that help blind and low-vision users consume and produce web content on unknown desktop user interfaces via audio spatialization and directional navigation
- Created a mobile application with React Native and Socket.io for a 20-person study that discovered new categories of assistive technologies that can create connection between sighted and blind and low-vision users in exploration

University of Washington Tacoma

REU Participant | Advisor: Jim Thatcher

2021

Tacoma, WA

- Used tools of computational topology, statistics, and machine learning to discover communities of interest for fair representation in political redistricting and better legislation for marginalized groups in the United States
- Developed data pipelines using Python, Pandas, and NumPy to specify parameters for algorithms, add flags to imported datasets based on block group clusters, and calculated summary statistics for each detected political district

WORK EXPERIENCE

Capital One

Software Engineer

2023 – 2026

McLean, VA

- Developed custom MITRE-based security heatmap using Angular and React to surface vulnerability gaps, implementing saved layer panels and custom styling for streamlined threat discovery across the enterprise
- Built first iteration of in-house monitoring system using React, TypeScript, and Python to replace Grafana dashboards, implementing critical cloud asset visibility metrics for platform admins
- Implemented dynamic memory allocation for S3 processing using Python, automatically scaling resources based on dataset size—saving \$10K+ monthly and preventing over-provisioning across 300+ datasets
- Upgraded file upload tool by translating Python validation script into Java and integrating real-time error feedback with Angular—slashing processing time by 2 hours and eliminating all duplicate uploads
- Spearheaded redesign of toxic access pairs plugin, resulting in a 40% reduction in security vulnerabilities and 25% increase in compliance with access control policies
- Led frontend development of AWS secrets management plugin using AngularJS and Java, reducing configuration time by 75% across 200 AWS accounts and streamlining vault operations for engineering teams

CONFERENCE PAPERS & PREPRINTS

Ruanqianqian (Lisa) Huang, **Avery Reyna**, Sorin Lerner, Haijun Xia, and Brian Hempel. “Professional Software Developers Don’t Vibe, They Control: AI Agent Use for Coding in 2025.” *arXiv Preprint*.

Yuanyang Teng, Connor Courtien, David Rios, Yves M. Tseng, Jacqueline Gibson, Maryam Aziz, **Avery Reyna**, Rajan Vaish, and Brian Smith. “Help Supporters: Exploring the Design Space of Assistive Technologies to Support Face-to-Face Help Between Blind and Sighted Strangers.” *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*.

Arnavi Chheda-Kothary, David Rios, Kynneddy Simone Smith, **Avery Reyna**, Cecilia Zhang, and Brian Smith. “Understanding Blind and Low Vision Users’ Attitudes Towards Spatial Interactions in Desktop Screen Readers.” *The 25th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '23)*.

PUBLIC SCHOLARSHIP

Avery Reyna. “The Threat of Artificial Intelligence Is Not Just Real, It’s Here.” March 29, 2023. *The Loop*.

Avery Reyna. “Interdisciplinary Social Science and the Limits of Quantitative Research.” October 25, 2022. *The Loop*.

Salah Ben Hammou and **Avery Reyna.** “Anti-Coup Strategies Should Address Civilian Coup Allies.” July 20, 2022. *Just Security*.

Avery Reyna and Salah Ben Hammou. “Reflecting on Coup Risk in Mali.” June 14, 2022. *Political Violence at a Glance*.

Avery Reyna. “Integrated Data in the United States: A Look at New York’s Workforce Portal.” May 13, 2022. *New America*.

Avery Reyna. “Latin America’s Vaccination Efforts: What to Know.” May 20, 2022. *Council on Foreign Relations*.

TALKS

Alexis Wood, **Avery Reyna**, Aya Konishi, David Retchless, and Jim Thatcher. “Cartographic Limits and Interdisciplinary Science: Maps and Mathematics in Pursuit of Social Justice.” *North American Cartographic Information Society (NACIS '22)*.

Jonathan Powell and **Avery Reyna.** “Legacies of Instability: Military Coups and Regime Consolidation in the Modern World.” *International Studies Association South (ISA South '22)*.

PRESENTATIONS

Avery Reyna and Jonathan Powell. “Applied Machine Learning to Detect Risk of Illegal Leadership Turnover.” *UCF Summer Undergraduate Research Poster Showcase '22*.

Avery Reyna, Ashwaq Alsoubai, and Pamela Wisniewski. “Comparative Analysis between TextRank and LexRank: Testing Utility of Text Summarization Models for Data Annotation.” *UCF Student Research Week '22*.

Carlson Sharpless, Komila Khamidova, **Avery Reyna**, Afsaneh Razi, and Pamela Wisniewski. “Adolescent Online Safety: Expanding the Knowledge Base and Developing Machine Learning Algorithms.” *UCF Student Research Week '21*.

AWARDS & HONORS

Senior Scholar – UCF School of Politics, Security, and International Affairs	2023
Summer Undergraduate Research Fellowship – UCF Office of Undergraduate Research	2022
Political Science Endowed Scholarship – UCF School of Politics, Security, and International Affairs	2020
Leader-in-Residence, Hispanic Association of Colleges and Universities	2020
Excellence in Action Award – UCF Office of Diversity Education and Training	2020
Burnett Honors Scholar – Burnett Honors College	2019